

King Fahad Central Hospital

Acute Stroke CT Imaging Protocol

Non-Contrast CT, CT Angiography & CT Perfusion - Aligned with AHA/ASA Guidelines and Target: Stroke Imaging Metrics

1. Purpose & Imaging Time Targets

Rapid, protocol-driven neuroimaging is central to acute stroke triage: it excludes hemorrhage, estimates infarct burden, identifies large vessel occlusion (LVO), and - in the extended time window - characterizes the ischemic core/penumbra to guide reperfusion therapy decisions.

Metric	Target
Door-to-CT initiation (non-contrast)	? 25 minutes from ED arrival
Door-to-CT interpretation	? 45 minutes from ED arrival
CT table notification	Stroke alert/code stroke activation triggers direct-to-scanner notification, bypassing routine queue

Targets reflect commonly cited AHA/ASA Target: Stroke initiative benchmarks - confirm current benchmark values against the active guideline/registry version.

2. Activation Workflow

- [] Stroke alert/code stroke activated by ED per LKW time and clinical screen - CT/CT technologist notified simultaneously with stroke team
- [] Patient transported directly to CT table on arrival, bypassing routine registration/imaging queue when protocol allows
- [] Stroke order set imaging orders (non-contrast CT +/- CTA +/- CTP) placed as a single bundled protocol to avoid sequential ordering delays
- [] Weight, renal function (if available), and contrast allergy history communicated to CT technologist before contrast administration
- [] IV access confirmed adequate for power injection (contrast studies) before patient reaches scanner

3. Non-Contrast CT Head

3.1 Purpose

- [] Exclude intracranial hemorrhage (absolute requirement before thrombolysis)
- [] Assess early ischemic changes and estimate infarct core burden via ASPECTS scoring
- [] Identify alternative diagnoses (mass, prior infarct, other structural lesion)

3.2 Acquisition (parameters set by Radiology/CT physics per scanner)

- [] Thin-section axial acquisition from skull base through vertex, reconstructed to standard soft-tissue and bone algorithm
- [] No IV contrast for this initial sequence
- [] Reconstruction slice thickness per departmental stroke protocol (typically thin sections for ASPECTS scoring plus standard thicker sections for routine read)

3.3 Interpretation - ASPECTS

Alberta Stroke Program Early CT Score: 10-point scoring system evaluating MCA territory; 1 point subtracted for each of 10 defined regions showing early ischemic change (score of 10 = no early changes; lower scores indicate larger territory involved).

- [] ASPECTS documented in the read and communicated verbally to the stroke team for time-critical decisions
- [] ASPECTS ?6 is a standard eligibility criterion referenced in early-window thrombectomy selection (see institutional Mechanical Thrombectomy Selection Protocol)
- [] Hemorrhage, mass effect, or established large completed infarct documented and communicated immediately - may preclude thrombolysis/thrombectomy

4. CT Angiography (Head & Neck)

4.1 Purpose

- [] Identify and localize large vessel occlusion (intracranial ICA, MCA-M1/M2, basilar, vertebral)
- [] Assess collateral circulation status
- [] Evaluate cervical vasculature for tandem occlusion, dissection, or significant carotid stenosis relevant to intervention planning

4.2 Acquisition

- [] Coverage from aortic arch (or proximal great vessels per departmental protocol) through vertex

[] IV contrast bolus via power injector through adequate-gauge peripheral IV, timed to arterial phase (bolus tracking or fixed-delay technique per departmental protocol)

[] Single-phase CTA is standard; multiphase CTA (arterial, peak venous, late venous) may be used per departmental protocol to further characterize collateral status

[] Thin-section reconstruction with multiplanar and maximum-intensity-projection (MIP) reformats for vessel evaluation

4.3 Contrast Safety Screening

[] Renal function (creatinine/eGFR) reviewed per departmental contrast policy - do not delay emergent CTA in a thrombectomy candidate solely pending renal function unless departmental policy specifies otherwise; follow institutional emergent-contrast pathway

[] Contrast allergy history reviewed; premedication per departmental protocol if history present and time allows, or per emergent allergy pathway

[] Metformin use noted per departmental post-contrast renal monitoring policy

[] IV hydration per departmental protocol if renal impairment identified and time permits

5. CT Perfusion (CTP)

5.1 Purpose

[] Estimate ischemic core (irreversibly infarcted tissue) and penumbra (at-risk but potentially salvageable tissue) volumes

[] Support core/penumbra mismatch-based patient selection for thrombectomy in the extended time window (commonly 6-24h from LKW) per DAWN/DEFUSE 3-based criteria

[] Not required for early-window (0-6h) candidates who meet standard non-contrast CT/CTA criteria, per institutional selection protocol

5.2 Acquisition

[] Dynamic cine acquisition through the region of interest during IV contrast bolus, per departmental CTP protocol and scanner-specific technique

[] Post-processed using validated automated software (e.g., RAPID, Viz.ai, or institutional equivalent) to generate CBF, CBV, MTT/Tmax maps

[] Automated core and penumbra volumes, and mismatch ratio, generated and transmitted to stroke team per departmental workflow

5.3 Key Perfusion Parameters

8. Quality Metrics to Track

- ☐ Door-to-CT (needle) initiation time
- ☐ Door-to-CT interpretation/verbal communication time
- ☐ Door-to-CTA completion time (for LVO detection)
- ☐ Imaging-to-groin-puncture time for thrombectomy patients
- ☐ Rate of contrast-related adverse events

Radiology / Stroke Team Sign-Off

Radiologist Signature: _____ Date/Time: _____

CT Technologist: _____ Stroke Team Notified (name/time): _____

References: 2019 AHA/ASA Guidelines for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates; AHA/ASA Target: Stroke initiative imaging time benchmarks; AHA/ASA Get With The Guidelines-Stroke measures; DAWN trial (Nogueira et al., NEJM 2018) and DEFUSE 3 trial (Albers et al., NEJM 2018) perfusion-based selection criteria; The Joint Commission Primary/Comprehensive Stroke Center imaging turnaround requirements. This template must be re-verified against current departmental scanner protocols, contrast policies, and the current guideline version at the time of institutional adoption.